

CSE HACKATHON 2026

1. AI Multi-Agent Campus Emergency Response & Resource Coordination System

Domain: College / Campus Safety / Disaster Management / Agentic AI

Problem Statement: Universities and large campuses face fire incidents, medical emergencies, security incidents, severe weather, accidents and crowd-related events. Information is distributed across security, medical, transport, facilities and administration teams, making rapid coordination difficult.

Objective: Develop a multi-agent emergency command system that understands an incident, determines its severity and location, identifies available campus resources, recommends response actions and coordinates authorized response teams through a unified dashboard.

Expected Prototype Features

- Incident intake through text, voice, image or structured reports.
- AI incident classification, summarization and severity assessment.
- Specialized agents for security, medical response, transport, facilities and communication.
- Campus-map-based safe route and resource recommendations.
- Availability tracking for ambulances, security personnel, first-aid units, shelters and campus vehicles.
- Dynamic re-planning when resources or conditions change.
- Emergency briefings, alerts and incident reports.
- Human approval and audit trail for high-impact actions.

2. Decentralized Graph-RAG Academic Advising & Prerequisite Conflict Resolver

Domain: College / Academic Intelligence / Graph AI / Agentic AI

Problem Statement: Academic planning becomes difficult with interdisciplinary courses, transferred credits, complex prerequisites, changing curricula and different career goals. Conventional portals do not reason across the complete academic pathway.

Objective: Build an agentic academic advisor that understands curriculum rules, prerequisites, credits, student history and career objectives to generate valid semester-wise pathways and identify academic bottlenecks.

Expected Prototype Features

- Knowledge graph for courses, prerequisites, credits and curriculum rules.
- Graph-RAG retrieval of academic policies and course information.
- Student-specific degree pathway generation.
- Prerequisite and credit conflict detection.
- Graduation-risk and course-bottleneck identification.
- Alternative course or substitution recommendations.
- Citation-traceable natural-language advising.
- Formal constraint verification and faculty approval for exceptional cases.

3. AI Campus Event Planning & Coordination Agent

Domain: College / Campus Operations / Multi-Agent AI

Problem Statement: Technical fests, conferences, workshops, hackathons and placement drives require coordination of venues, schedules, equipment, volunteers, security, transport, communication and permissions.

Objective: Develop an agentic event-planning platform that converts event requirements into an operational plan, checks constraints, coordinates tasks and dynamically replans when conditions change.

Expected Prototype Features

- Natural-language event requirement intake.
- Venue, equipment, volunteer and support-team planning.
- Schedule and resource conflict detection.
- Alternative venue and schedule recommendations.
- Task delegation and deadline tracking.
- Event-readiness dashboard and automated checklists.
- Notifications and stakeholder briefings.
- Human approval for permissions, budget and security-sensitive actions.

4. AI Student Attendance, Engagement & Intervention Agent

Domain: College / Education / Predictive AI / Agentic AI

Problem Statement: Attendance alone does not provide a complete view of student engagement. Faculty and mentors need to combine attendance, assessments, assignments, LMS activity and other authorized indicators.

Objective: Develop an AI agent that analyzes academic-engagement signals, detects meaningful changes, explains contributing factors and recommends personalized interventions to faculty mentors.

Expected Prototype Features

- Attendance, assessment, assignment and LMS integration.
- Engagement trend and anomaly detection.
- Explainable academic-risk profiles.
- Personalized intervention recommendations.
- Faculty mentor prioritization dashboard.
- Automated mentor summaries and action lists.
- Intervention tracking and outcome monitoring.
- Role-based access and privacy controls.

5. Multimodal Campus Emergency Early-Warning & Autonomous Crowd Evacuation

Router

Domain: College / Computer Vision / Safety / Smart Campus

Problem Statement: During campus emergencies, response can be delayed by incomplete situational awareness and changing crowd movement. Static evacuation plans may become unsafe when exits or routes are blocked.

Objective: Develop a multimodal campus safety system that detects abnormal crowd behavior and hazards from authorized camera feeds, estimates crowd density and dynamically recommends safe evacuation routes.

Expected Prototype Features

- Multi-camera crowd monitoring and anomaly detection.
- Crowd-density and movement analysis.
- Selected hazard and blocked-route detection.
- Dynamic hazard-zone mapping.
- Indoor positioning and campus routing simulation.
- Zone-based evacuation guidance.
- Emergency command dashboard.
- Offline or low-bandwidth evacuation guidance prototype.

6. AI Research & Project Collaboration Agent for M.Tech Students

Domain: College / Research Intelligence / GenAI / Knowledge Graphs

Problem Statement: M.Tech students spend considerable effort finding literature, organizing references, identifying research themes, locating datasets and tools, evaluating research gaps and converting ideas into feasible research plans.

Objective: Develop a research collaboration agent that takes a research idea and produces an evidence-backed research roadmap using literature, datasets, tools, methodologies and experiment planning.

Expected Prototype Features

- Semantic literature retrieval.
- Paper summarization and comparison.
- Research-theme and gap analysis.
- Dataset, benchmark and tool recommendations.
- Research methodology and experiment planning.
- Citation-aware research notes.
- Milestone and research-roadmap generation.
- Supervisor review and approval checkpoints.

7. AI Campus Placement Operations & Interview Coordination Agent

Domain: College / Placement / Employability / Agentic AI

Problem Statement: Campus placement requires eligibility checking, job-description analysis, candidate matching, interview scheduling, panel coordination, venue management and repeated communication.

Objective: Develop an AI placement operations agent that coordinates the workflow from company requirements through eligibility verification, candidate matching, interview scheduling and event reporting, while retaining human control over final selection.

Expected Prototype Features

- Job-description and eligibility extraction.
- Student eligibility verification.
- Skill-based candidate matching with explanations.
- Interview and test scheduling.
- Panel and room coordination.
- Student notifications and reminders.
- Placement dashboard with pending actions and exceptions.
- Skill-gap and placement-readiness analytics.

8. AI Smart Campus Energy Optimization Agent

Domain: College / Sustainability / IoT / Predictive AI

Problem Statement: Campus classrooms, laboratories, hostels and offices have varying occupancy and energy demand. HVAC, lighting and equipment may operate without considering schedules, weather and historical consumption.

Objective: Develop an AI energy optimization agent that forecasts demand, detects abnormal consumption and recommends operational strategies for reducing energy use while maintaining academic and safety requirements.

Expected Prototype Features

- Building and facility energy monitoring.
- Occupancy-aware consumption analysis.
- Energy-demand forecasting.
- Anomaly and inefficient-usage detection.
- HVAC and lighting recommendations.
- What-if energy-saving simulation.
- Building-level sustainability dashboard.
- Human approval and rollback for operational changes.

9. AI Infrastructure Inspection & Maintenance Prioritization Agent

Domain: Infrastructure / Computer Vision / Predictive AI

Problem Statement: Roads, buildings, bridges, drainage systems and public infrastructure require regular inspection. Visual evidence, reports, complaints and maintenance history may be fragmented.

Objective: Develop an AI inspection agent that combines visual evidence, reports, complaints and maintenance history to identify defects, rank risks and generate a prioritized maintenance plan for human review.

Expected Prototype Features

- Image-based defect detection.
- Multimodal analysis of images and reports.
- Risk and severity ranking.
- Location-based issue mapping.
- Maintenance-history retrieval.

- Resource-aware maintenance prioritization.
- Evidence-linked engineering reports.
- Inspection and maintenance analytics.

10. Autonomous Urban Pothole, Structural Defect & Micro-Flooding Lifecycle Engine

Domain: Smart City / Urban Infrastructure / GIS / Computer Vision

Problem Statement: Urban road and drainage maintenance is often reactive. Potholes, surface degradation and waterlogging can interact and create mobility and safety problems.

Objective: Develop a visual and geospatial lifecycle platform that detects road defects, estimates severity, correlates infrastructure conditions with water-flow patterns and prioritizes municipal maintenance work orders.

Expected Prototype Features

- Citizen or dashcam image intake.
- Road-defect detection and segmentation.
- Approximate depth/severity estimation.
- GIS-based defect mapping.
- Drainage and waterlogging hotspot correlation.
- Duplicate complaint detection.
- Maintenance priority scoring.
- Work-order and contractor workflow simulation.

11. AI Civic Complaint-to-Resolution Intelligence Platform

Domain: Governance / Civic Services / Agentic AI

Problem Statement: Citizen complaints involving roads, drainage, waste, water, streetlights and public facilities must be classified, prioritized, routed and followed up across departments.

Objective: Build an agentic civic-service platform that transforms citizen reports into structured cases, identifies responsible departments, determines priority, coordinates follow-up and provides transparent status tracking.

Expected Prototype Features

- Text, image and location-based complaint intake.
- Complaint classification and summarization.
- Duplicate complaint detection.
- Department and team routing.
- Impact- and urgency-based prioritization.
- Escalation of delayed cases.
- Citizen status and resolution explanation.
- Recurring-problem analytics.

12. AI Healthcare Patient Journey & Follow-Up Coordination Agent

Domain: Healthcare / Patient Coordination / Multi-Agent AI

Problem Statement: Patients may interact with registration, consultation, diagnostics, pharmacy, specialist referrals and follow-up services. Fragmented administrative information can cause delays and missed follow-ups.

Objective: Develop a healthcare coordination agent that manages authorized administrative and care-coordination information across appointments, referrals, investigations and follow-ups, with human review for clinical decisions.

Expected Prototype Features

- Appointment coordination.
- Referral tracking.
- Investigation and follow-up monitoring.
- Patient journey timeline.
- Missing-action and overdue-follow-up detection.
- Multi-agent coordination across hospital services.
- Grounded summaries for authorized staff.
- Human review before clinically significant actions.

13. AI Smart Agriculture Decision Agent

Domain: Agriculture / Multimodal AI / Agentic AI

Problem Statement: Farm decisions depend on crop condition, soil, weather, irrigation, pest or disease risk and market conditions. Farmers often receive fragmented information rather than contextual decision support.

Objective: Develop a multi-agent agriculture decision system that combines crop images, soil information, weather, crop stage, irrigation and market information to provide explainable farm-management recommendations.

Expected Prototype Features

- Crop and field profile management.
- Crop-image analysis.
- Weather-aware recommendations.
- Soil and irrigation analysis.
- Disease and pest-risk assessment.
- Market and price intelligence.
- Integrated farm decision generation.
- Explainable recommendations and farmer feedback.

14. AI Water, Energy & Environmental Sustainability Optimization Agent

Domain: Sustainability / Smart Campus / Smart City

Problem Statement: Organizations and communities consume water and energy while generating waste and environmental impact. Resource anomalies and inefficient operational patterns may remain unnoticed without integrated intelligence.

Objective: Build an AI sustainability agent that analyzes energy, water, waste, air-quality, occupancy and facility data to identify anomalies, forecast demand and recommend coordinated sustainability actions.

Expected Prototype Features

- Energy and water forecasting.
- Anomaly and leakage-pattern detection.
- Occupancy-aware resource analysis.
- Facility benchmarking.
- AI-generated conservation recommendations.
- What-if analysis of proposed interventions.
- Environmental impact and savings tracking.
- Human approval before operational changes.

15. AI Community Resilience & Emergency Resource Coordination Agent

Domain: Community / Disaster Management / Multi-Agent AI

Problem Statement: Floods, cyclones, fires, heatwaves and other emergencies require simultaneous coordination of affected residents, shelters, healthcare, transport, food, water, volunteers, roads and infrastructure.

Objective: Develop a community resilience platform in which specialized agents collaboratively understand an emergency, identify needs, match resources, recommend priorities and continuously re-plan as conditions change.

Expected Prototype Features

- Emergency incident intake from citizen reports and simulated sensors.
- Affected-area and priority estimation.
- Resident-needs and resource matching.
- Shelter, healthcare, transport and volunteer coordination.
- Dynamic route and resource recommendations.
- Multi-agent emergency command dashboard.
- Scenario simulation and what-if analysis.
- Human-in-the-loop control and auditable emergency actions.